Quotient word storage: compositional bridge

Literal quotient storage

Zero UInt64 array; Array.set at current shift; same word remainder loop

↓

Checked loop refinement

Decoded stored loop equals previous qloop for every fuel value

↓

Checked final trim and exact output

Canonical normalized quotient words; decoded exact factor equals wanted

↓

Checked allocation arithmetic

Nonempty divisor + representable input length gives safe +1 and in-range shifts

↓

Remaining native interfaces

Allocator / byte capacity / compiled Rust; nonmonic inversion and pegcd